

New Phytologist Supporting Information

Article title: Provenance and environment jointly shape early regeneration in fir and beech: Evidence from distributed participatory trials across Europe

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The following Supporting Information is available for this article:

Fig. S1 Climate chamber trays for fir (*Abies*) and beech (*Fagus*) germination.

Fig. S2 Phenological stages used to score germination and early seedling development.

Fig. S3 Cumulative germination as a function of growing degree-days in the field experiments.

Fig. S4 Fir (*Abies*) phenological development in climate chambers and field experiments.

Fig. S5 Beech (*Fagus*) phenological development in climate chambers and field experiments.

Fig. S6 Diagnostic plots for mixed-effects models of germination.

Fig. S7 Climate chamber germination trajectories across provenances.

Fig. S8 Effect of experimental cycle length on germination metrics.

Fig. S9 Correlations among germination metrics, geography, and seed traits.

Fig. S10 LASSO-selected environmental predictors ranked by their standardized regression coefficients for fir and beech.

Fig. S11 Germination rate as a function of geographic and environmental transfer distances.

Table S1 Information about the seed stands and origin certificates.

Table S2 Environmental covariates used for variable selection in the LASSO regression.

Table S3 Germination metrics across species, provenances, treatments, and experimental cycles.

Table S4 Wald tests for fixed effects retained in the final mixed-effects germination models.

Table S5 Generalized additive models (GAMs) of germination rate as a function of geographic and environmental transfer distances.

Methods S1 Protocols for establishing and monitoring the *MyGardenOfTrees* climate chamber

and field experiments.

Methods S2 Climate and soil data and variable selection using LASSO regression.

Methods S3 Description of a hidden Markov model of seed germination, phenological development, and mortality, including the inference and importance sampling scheme.

Dataset S1 XLSForms and associated media files used for standardized data collection in the *MyGardenOfTrees* citizen science project.

Figure S1: Examples of trays used in the climate chamber trials. Left side: fir (*Abies* spp.) provenances; right side: beech (*Fagus* spp.) provenances. Each tray contained 100 seeds. Photos: Johannes Alt and Mert Celik.

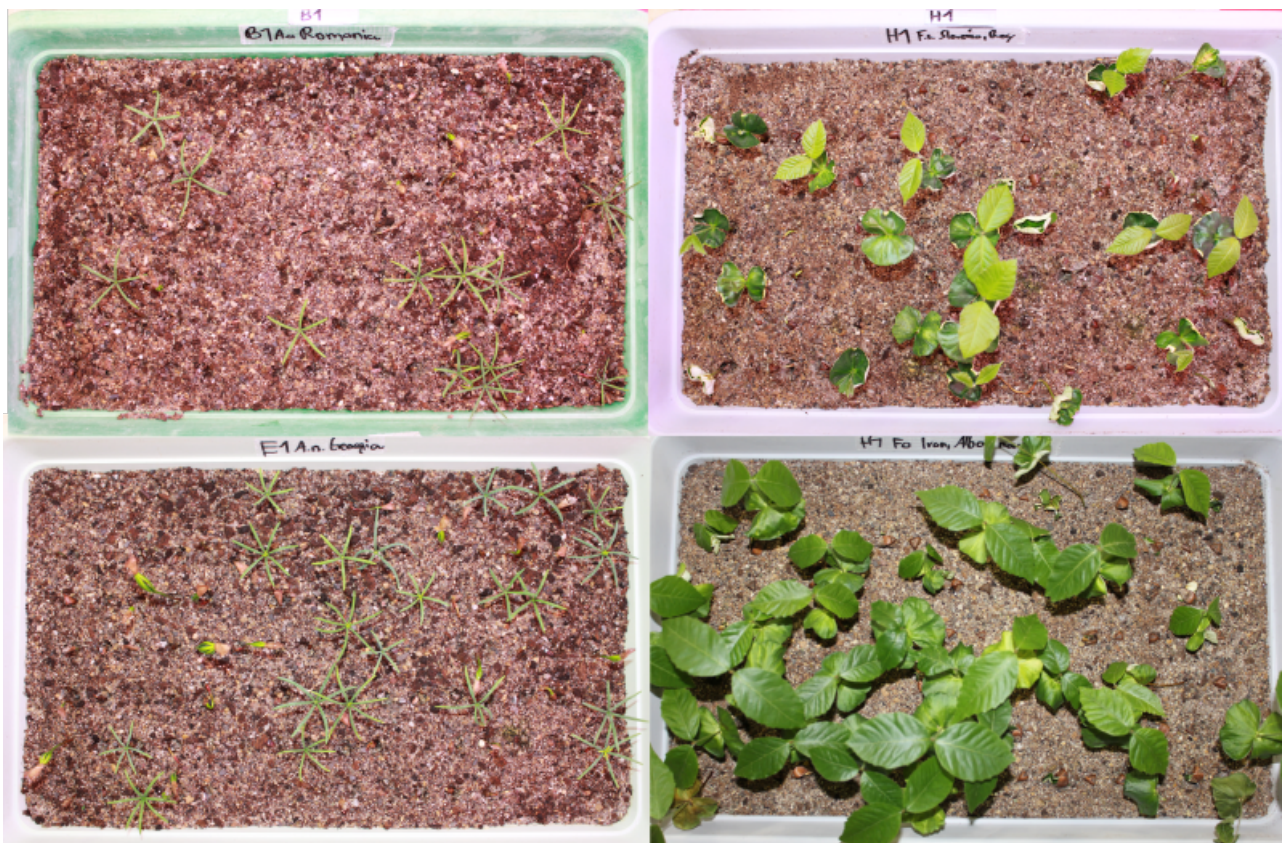


Figure S2: (a) Developmental stages of fir (*Abies*) seedlings. Stage 0: no germination is visible (not shown); stage 1: the seedling has emerged, there is a stem, but the seed coat still covers the needles; stage 2: the needles are visible, still grouped together or partially open, but the seed coat is still on; stage 3: the first set of needles is free from the seed coat, and the needles are open; stage 4: a second set of needles is visible. (b) Developmental stages of beech (*Fagus*) seedlings. Stage 0: no germination is visible (not shown); stage 1: the seed has emerged with a stem. The cotyledons are partially covered, and the seed coat is still on; stage 2: the seed coat has fallen; the cotyledons are still partially folded; stage 3: the cotyledons are open, and the first bud is visible between them, but the bud is still closed; stage 4: the first true leaves have emerged above the cotyledons but are still folded; stage 5: the first true leaves are completely unfolded. Photos: MyGardenOfTrees

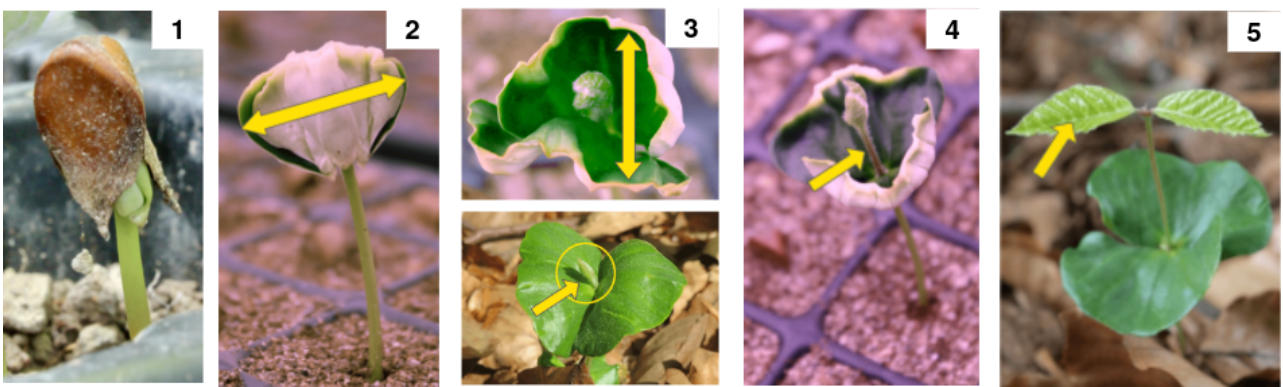
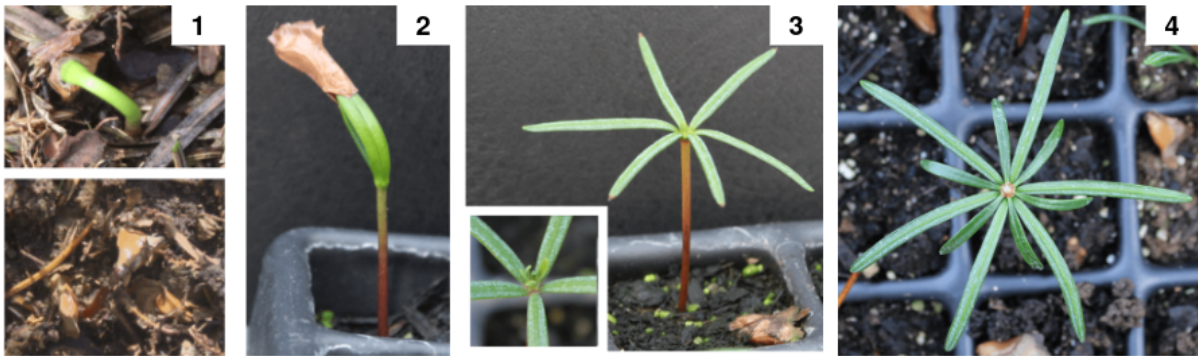
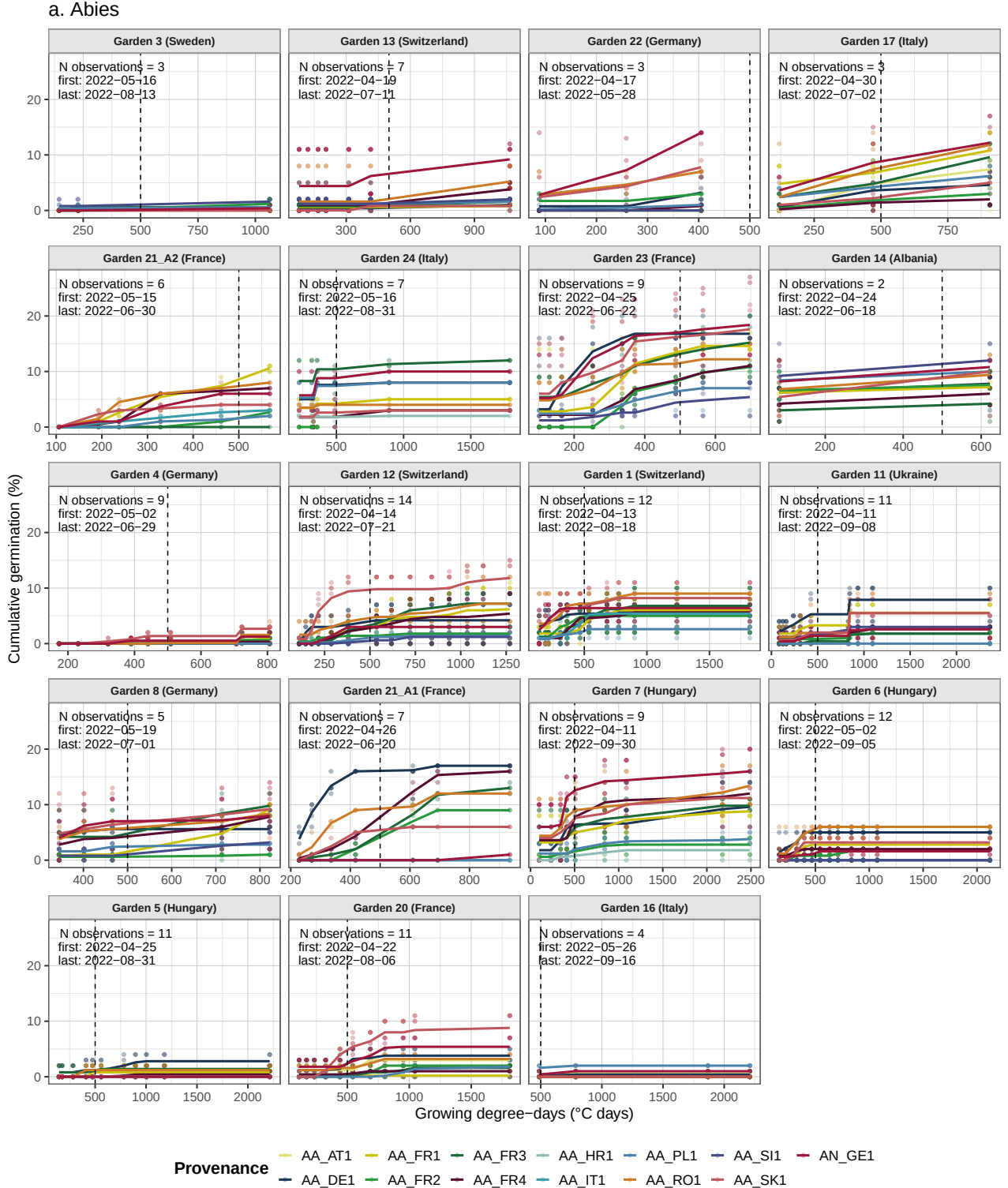


Figure S3: Cumulative germination (%) is shown as a function of growing degree-days (GDD) for all fir (*Abies*) (a) and beech (*Fagus*) (b) seed provenances in the first growing season. Each colored line represents one provenance, and panels correspond to gardens. Lines and points indicate observed cumulative germination at each census date, and smoothed monotonic curves illustrate overall germination trajectories. Panels are ordered by the mean GDD (from March to June) of each garden. Vertical dashed lines indicate GDD of 500 for each garden for comparison. Numbers in the upper-left corner of each panel indicate the number of observation dates used to construct the cumulative curves, as well as the first and last observations. Numbers in the upper-left corner of each panel indicate the number of observation dates used to construct the cumulative curves, as well as the first and last observations.

a. *Abies*



b. Fagus

b. Fagus

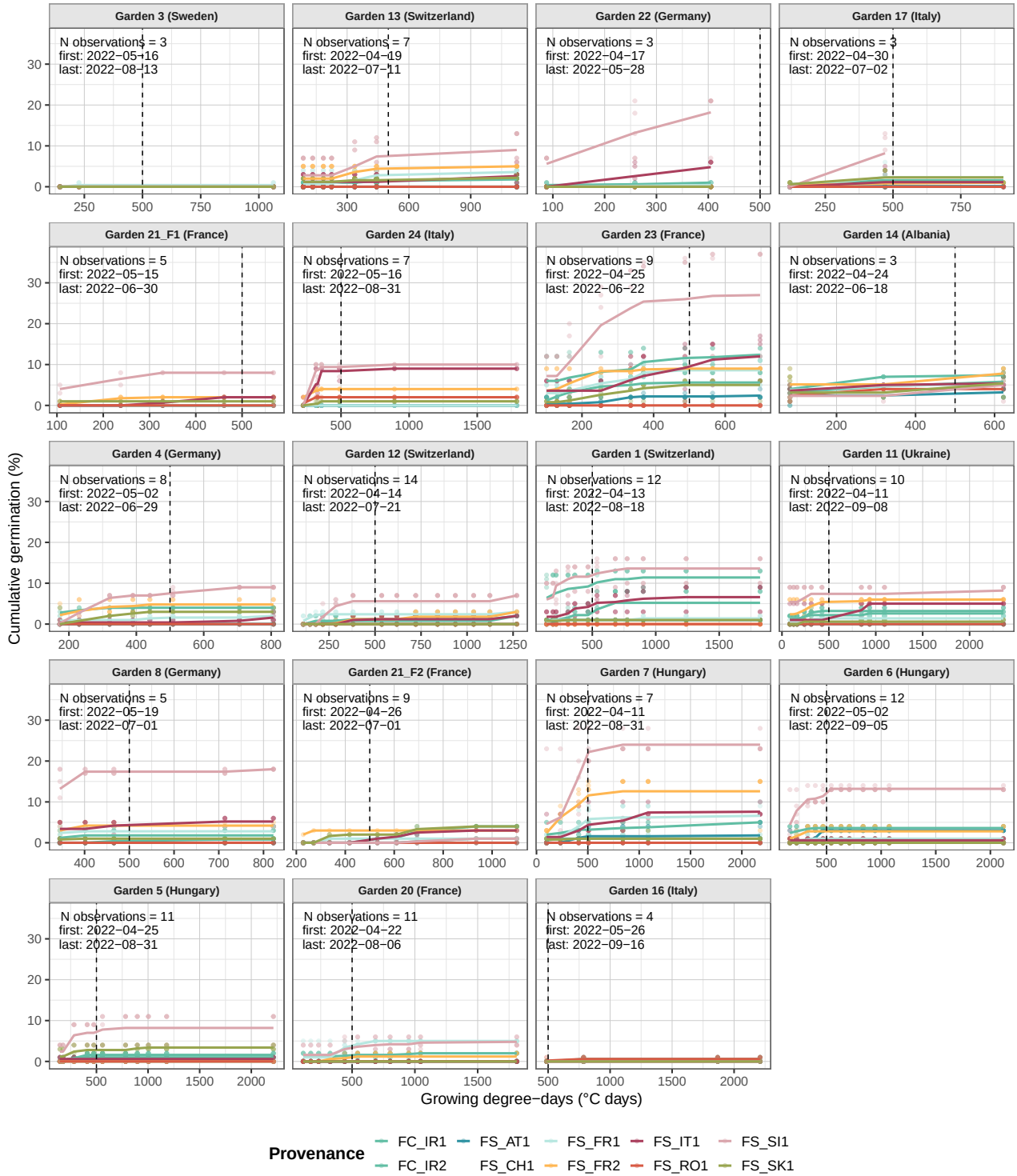


Figure S4: Phenological development of germinating fir (*Abies*) seeds in (a) the climate chambers and (b) the gardens. Stage 0 corresponds to ungerminated or dead seedlings. Note the difference in scale on the y-axis between (a) and (b).

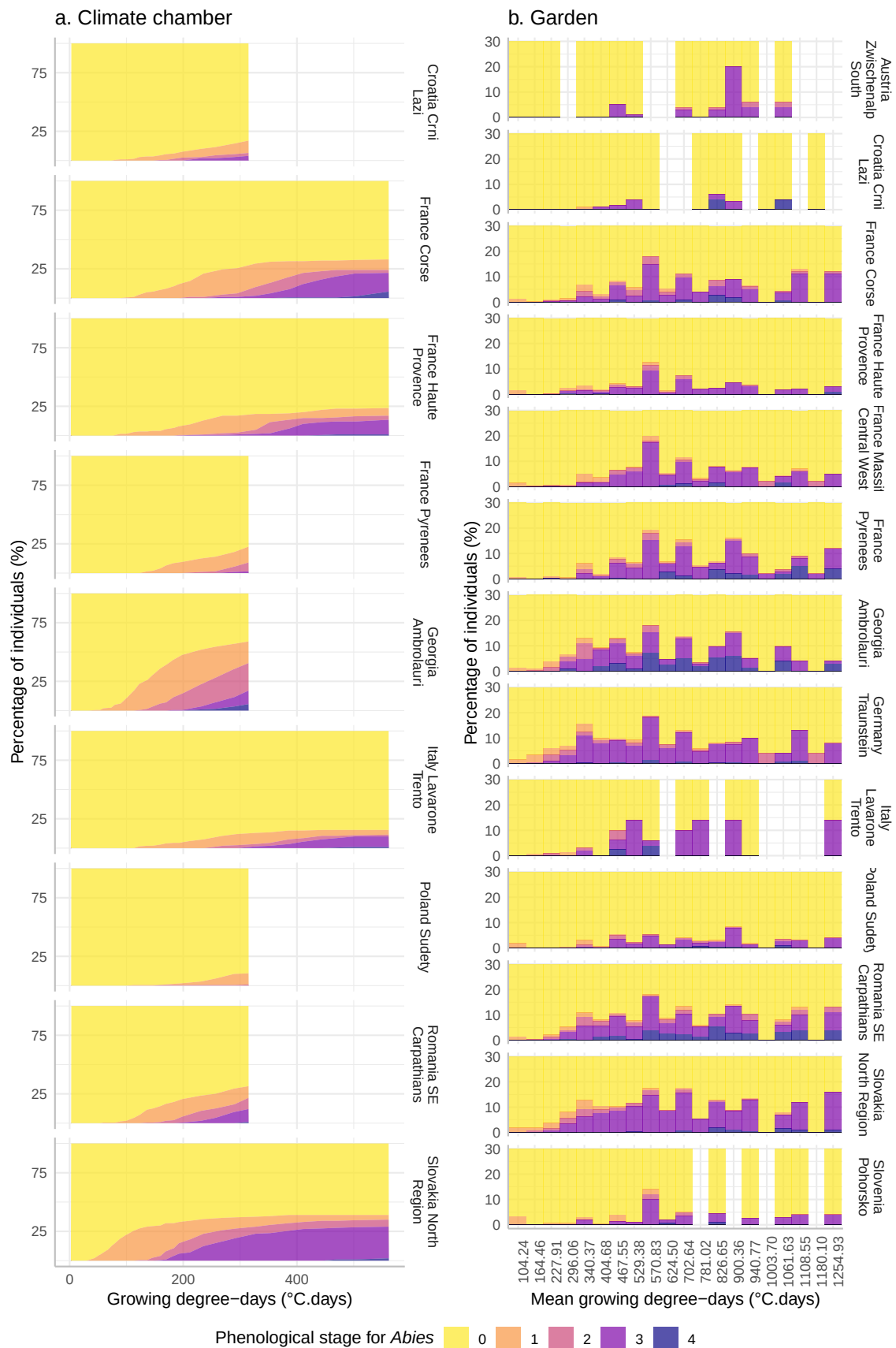


Figure S5: Phenological development of germinating beech (*Fagus*) seeds in (a) the climate chambers and (b) the gardens. Stage 0 corresponds to ungerminated or dead seedlings. Note the difference in scale on the y-axis between (a) and (b).

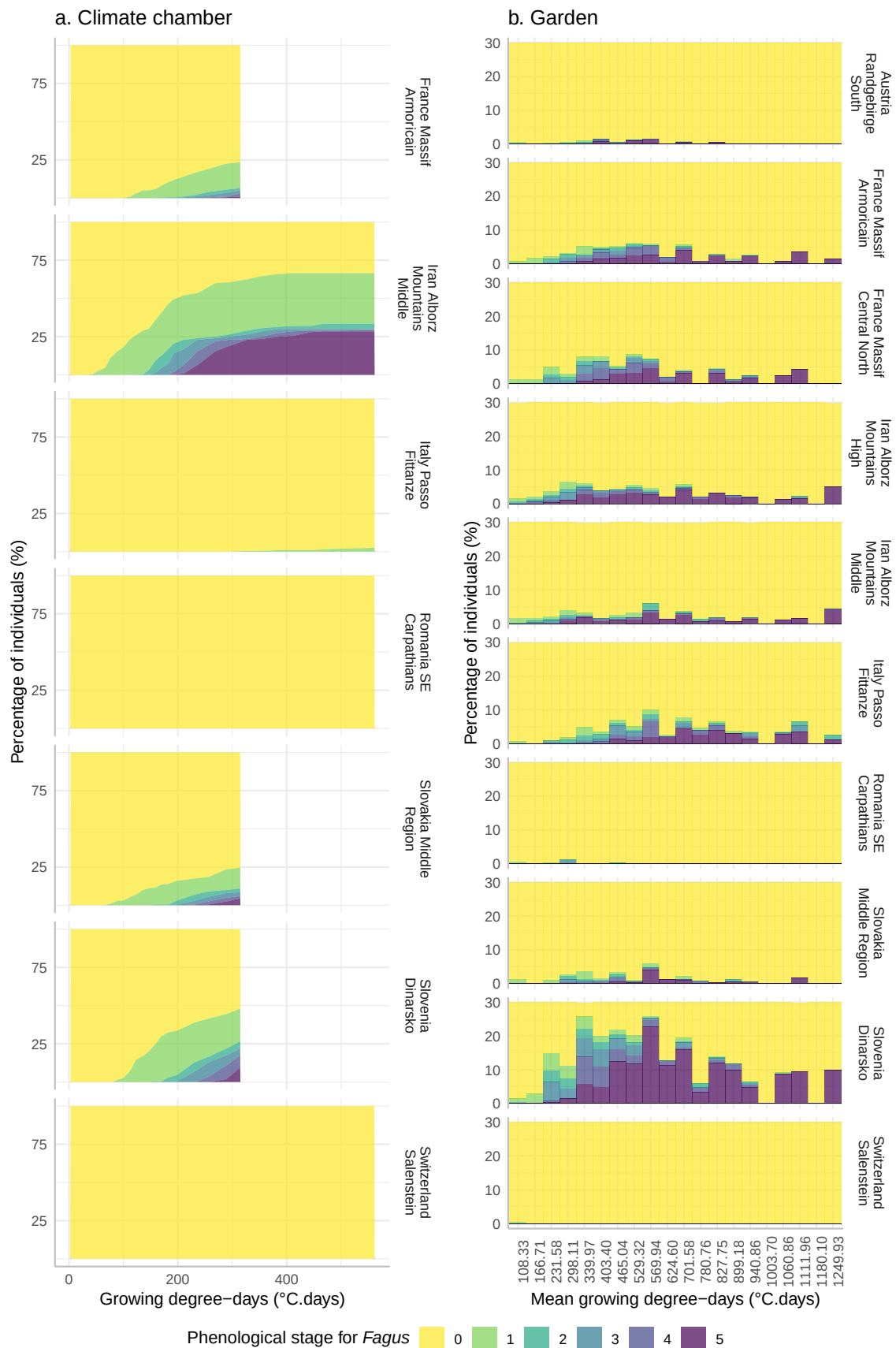
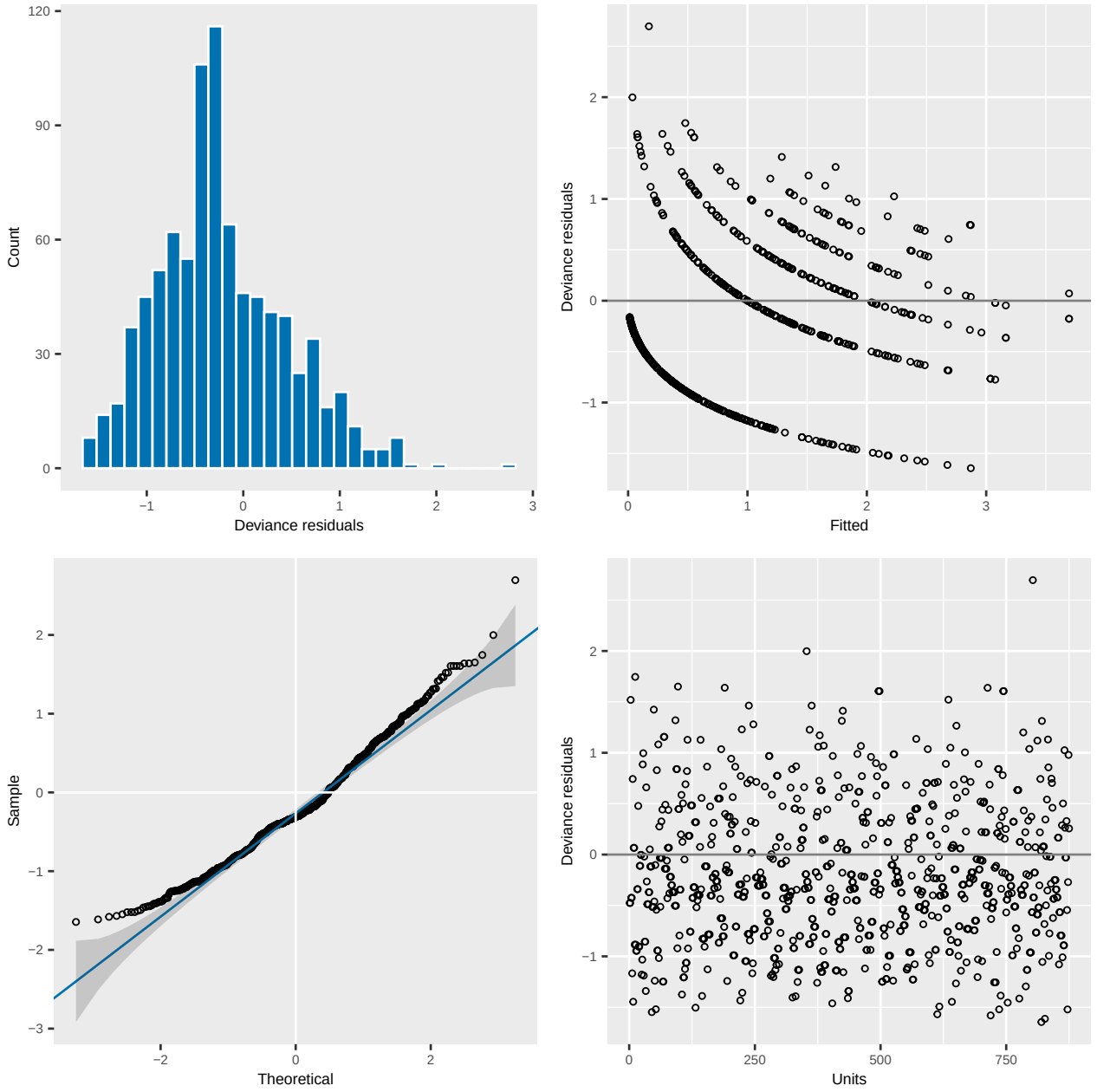


Figure S6: Diagnostic plots of the negative binomial model using ASReml Butler et al. [2023] for fir (a) and beech (b) species and provenances. The response variable is the cumulative maximum germination rate across all observation dates of a garden from the establishment of the experiment in winter 2022 until 30 June 2022. All fixed effects had a single unique value per garden block or provenance; no time-dependent predictors were included.

a. Abies



b. Fagus

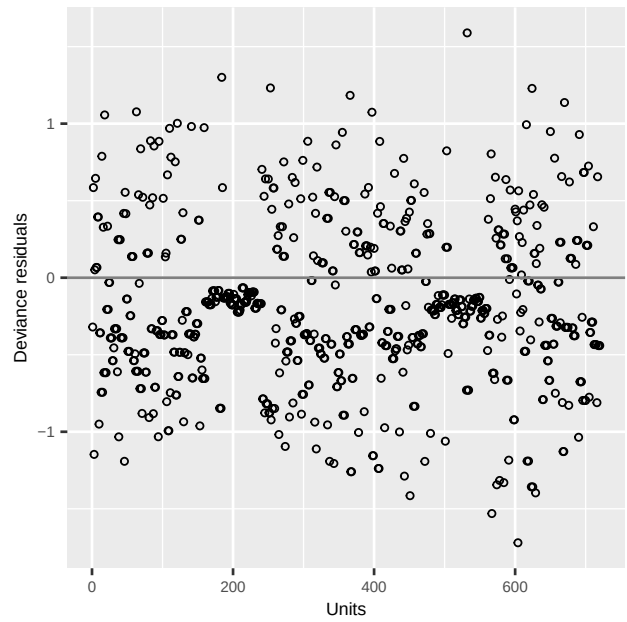
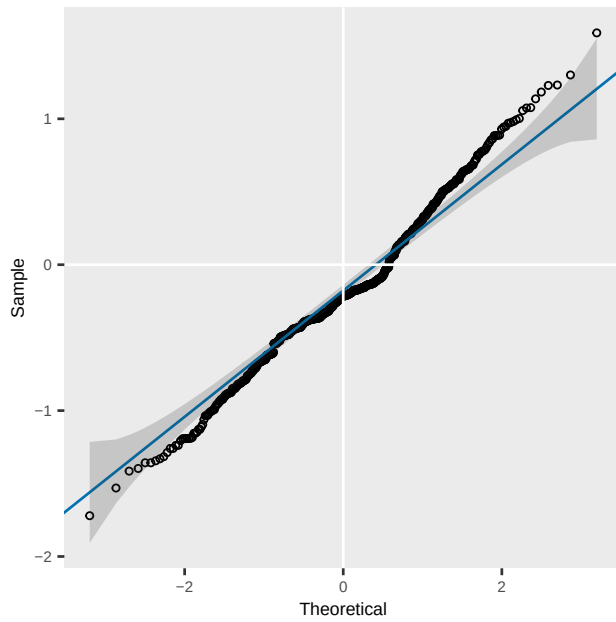
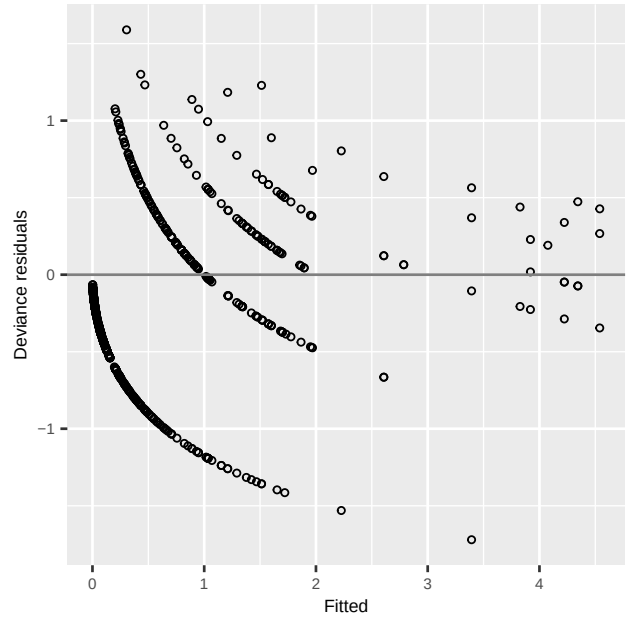
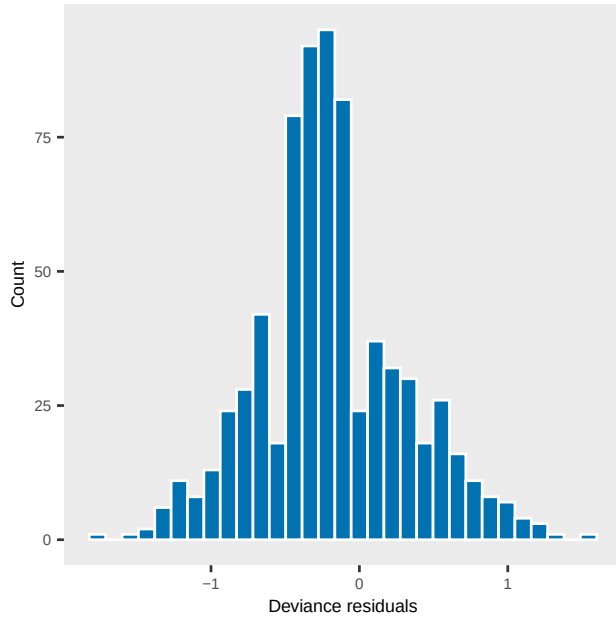


Figure S7: Germination in the climate chamber experiments for all studied provenances (Table 1). All fir provenances are *Abies alba*, but GEA1 (*A. nordmanniana*), and all beech provenances are *Fagus sylvatica*, but IRF1 (*F. caspica*). Provenances in the first cycle were observed for 72 days, while those in the second cycle were observed for 104 days. One day after the end of the cycles, seeds were dug up and inspected for roots that had been invisible during the observations; seeds with roots were added to the final count and marked with a closed symbol. Two provenances, ROF1 and CHF1, are excluded from the figure as no seeds germinated.

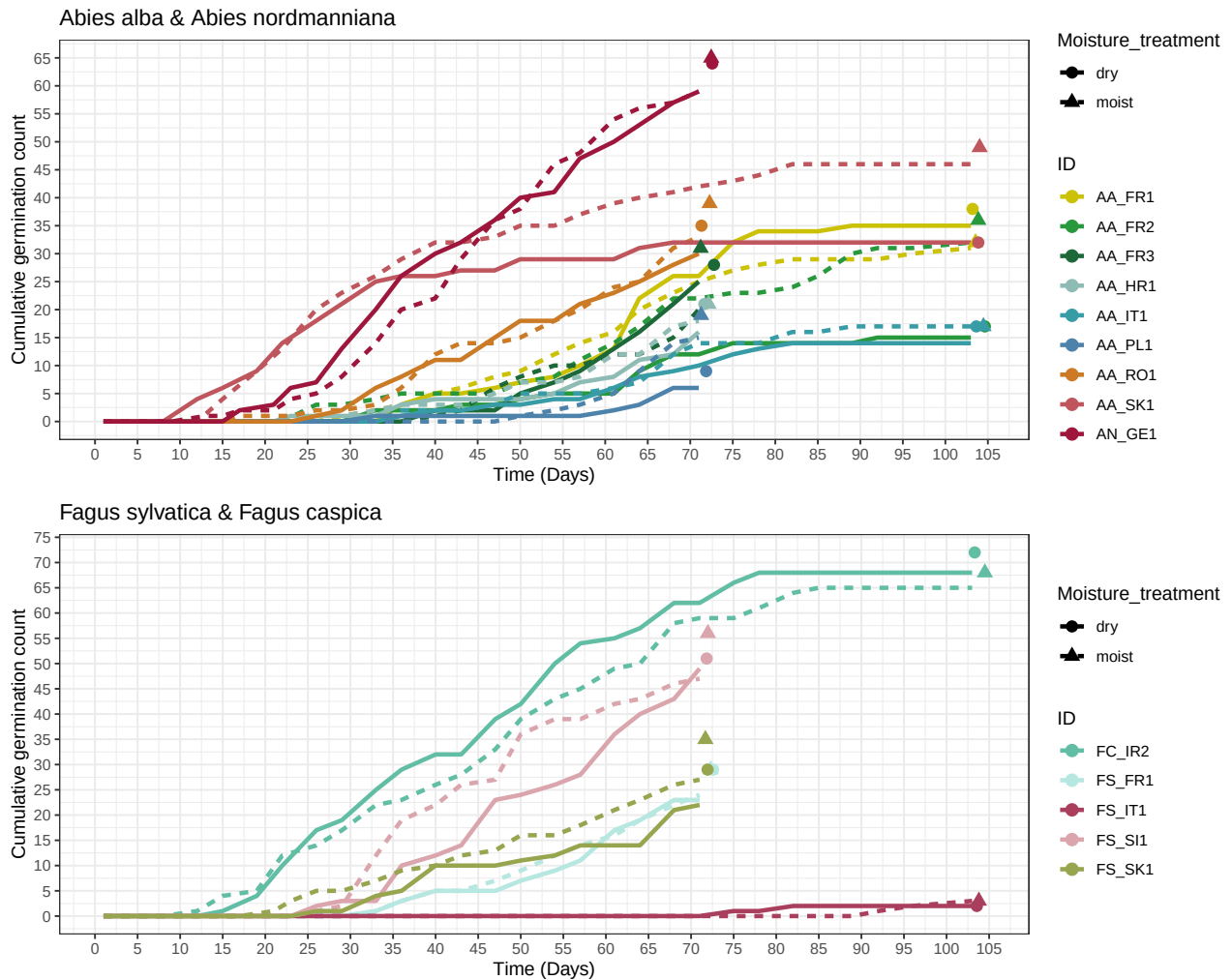


Figure S8: Effect of cycle length on the germination metrics.

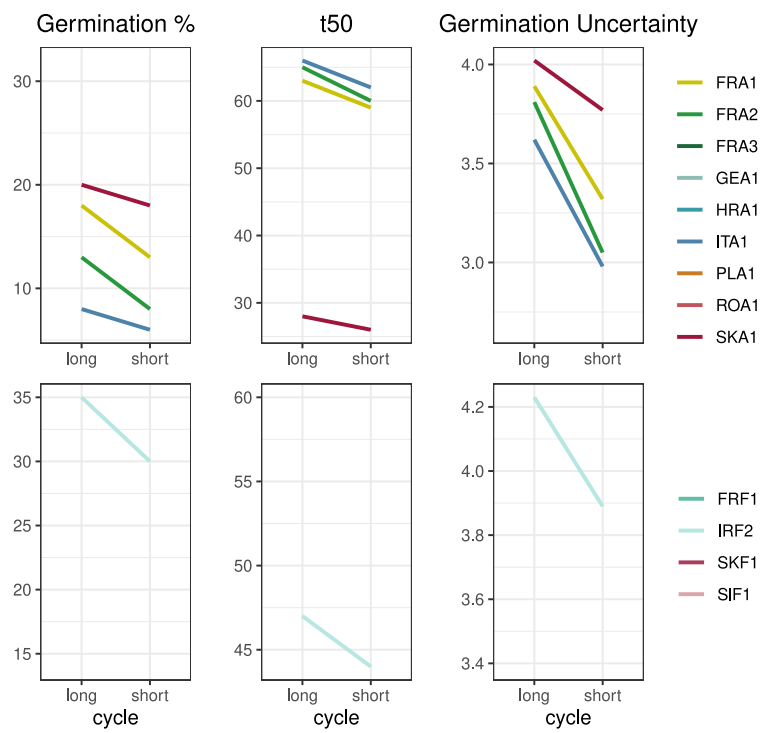
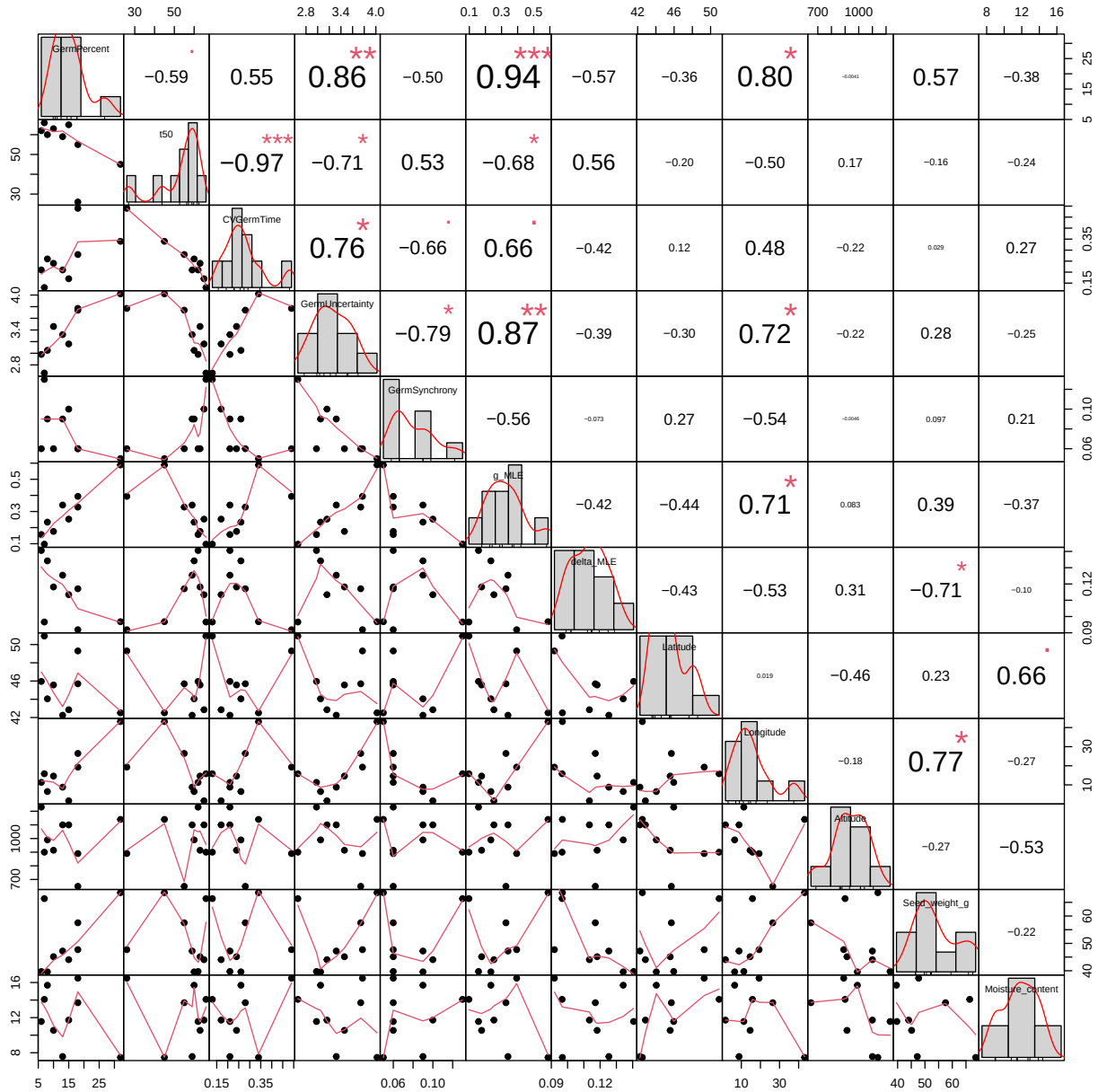
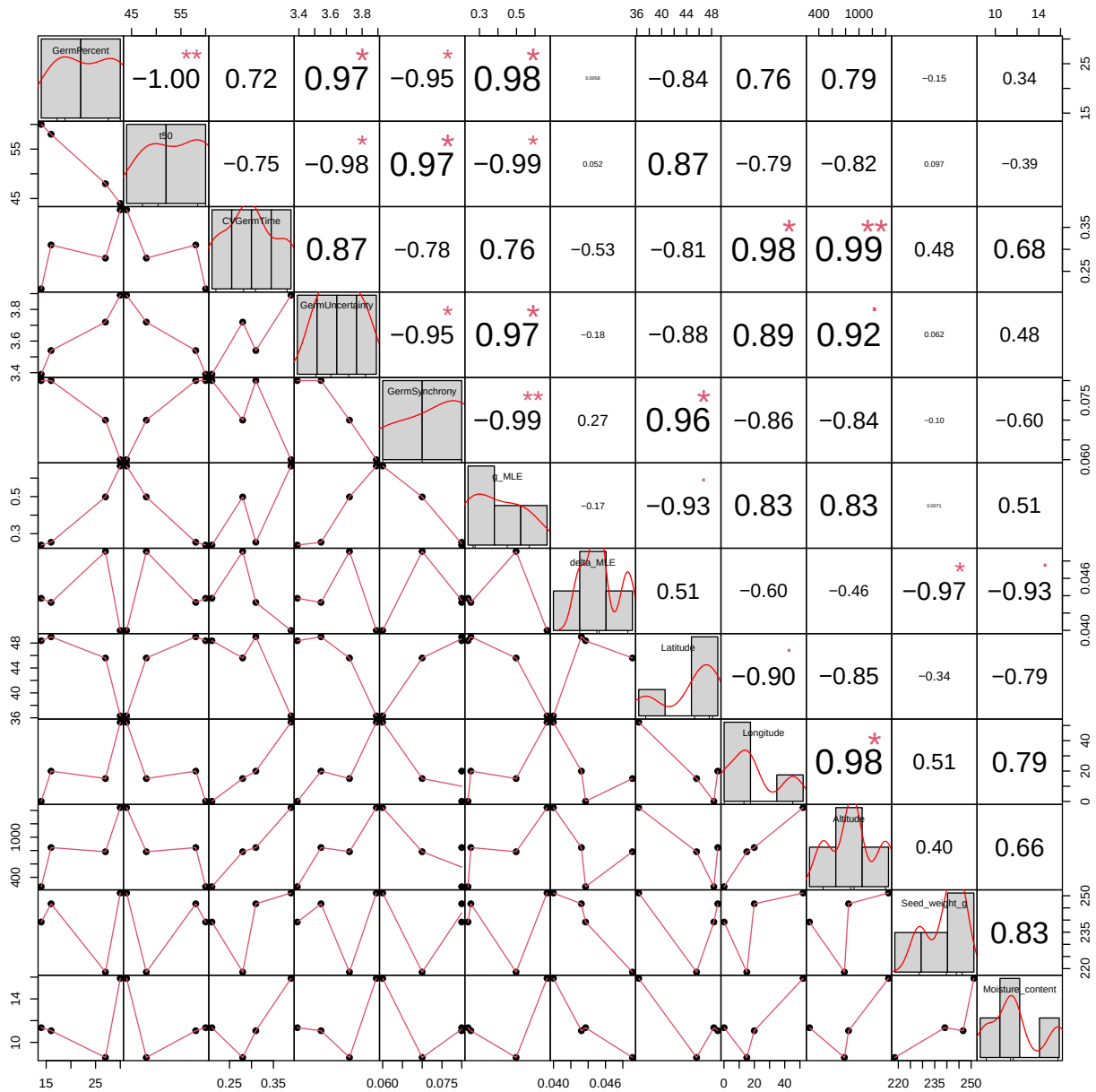


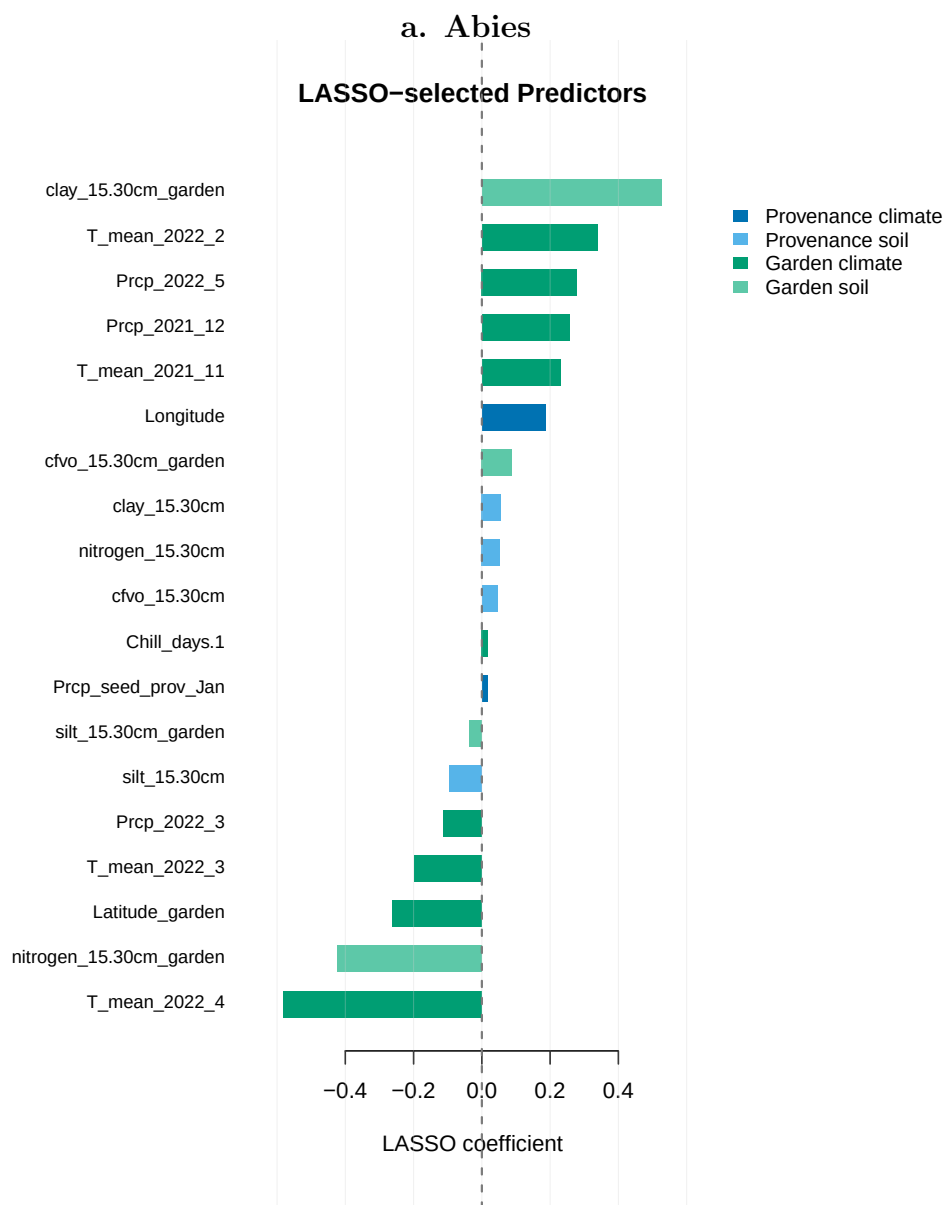
Figure S9: Raw data correlations between germination metrics, geography, and seed traits in fir (a) and beech (b) species and provenances in the climate chamber trials. See Materials and Methods for definitions.

a. Abies



b. Fagus





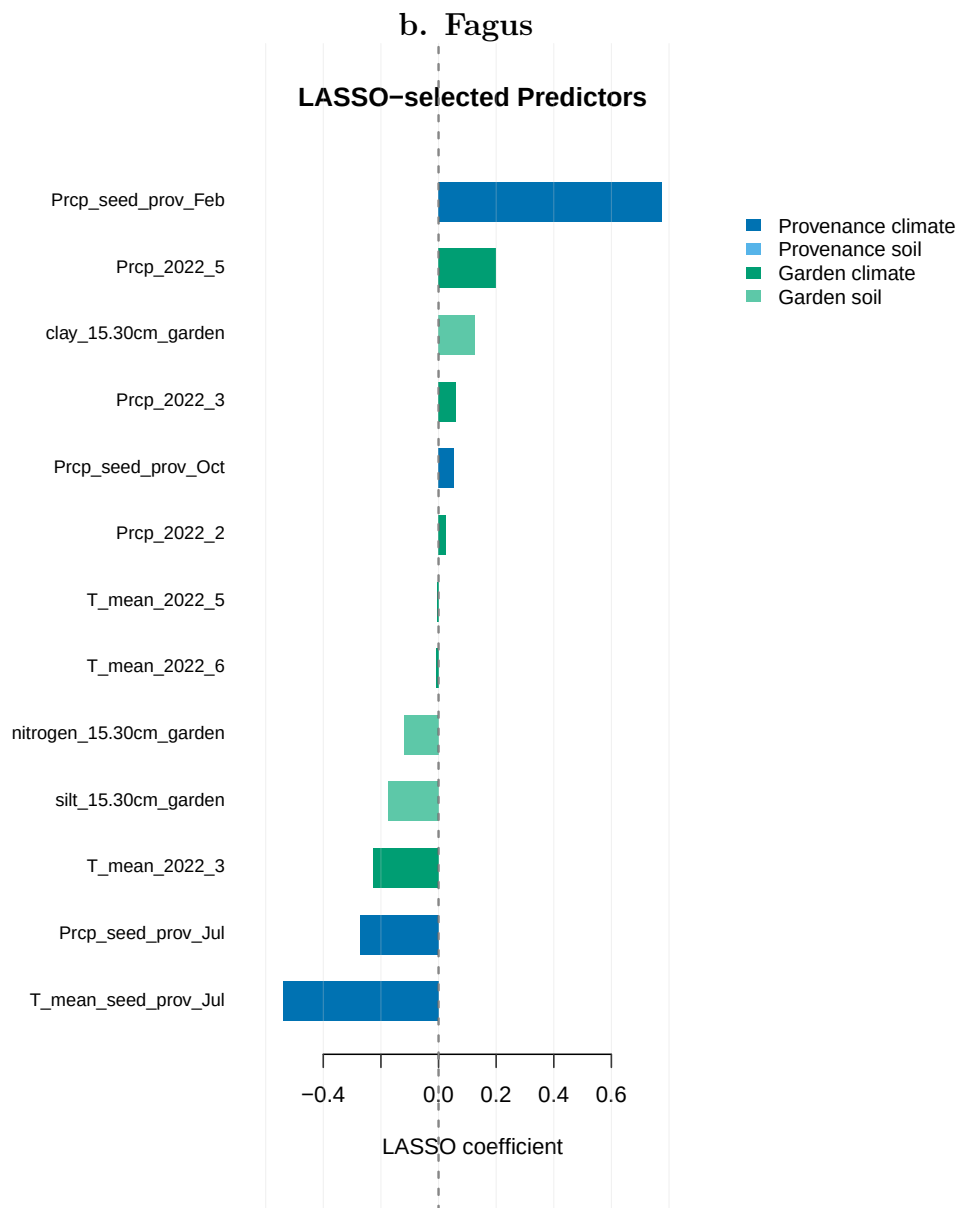


Figure S10: LASSO-selected environmental predictors ranked by their standardized regression coefficients for fir (*Abies*) (a) and beech (*Fagus*) (b) species. Horizontal bars show the direction and magnitude of the effect of each variable on cumulative germination probability, with negative values indicating reduced germination and positive values indicating increased germination. Variables were grouped (and colored) according to distinct types: seed provenance (long-term climatic conditions at the provenance site), micro-garden yearly (environmental variables with one unique value per year at the micro-gardens), provenance-by-micro-garden (climatic distances between provenance and micro-gardens), and micro-garden daily (environmental variables that change between observation dates).

Figure S11: Germination rate as a function of geographic and environmental transfer distances between provenance and garden, estimated from generalized additive models (GAMs) including provenance and garden identity as random effects. Positive values indicate transfer from environments with higher values than those experienced in the garden. Spring temperature and spring precipitation reflect the mean and total for March, April, and May. Points represent individual provenance–garden combinations and are colored by provenance. Solid lines show partial effects, dashed lines approximate 95% confidence intervals, and significance labels in the top-right corner correspond to the smooth terms of the transfer distance. See Table S5 for full model outputs.

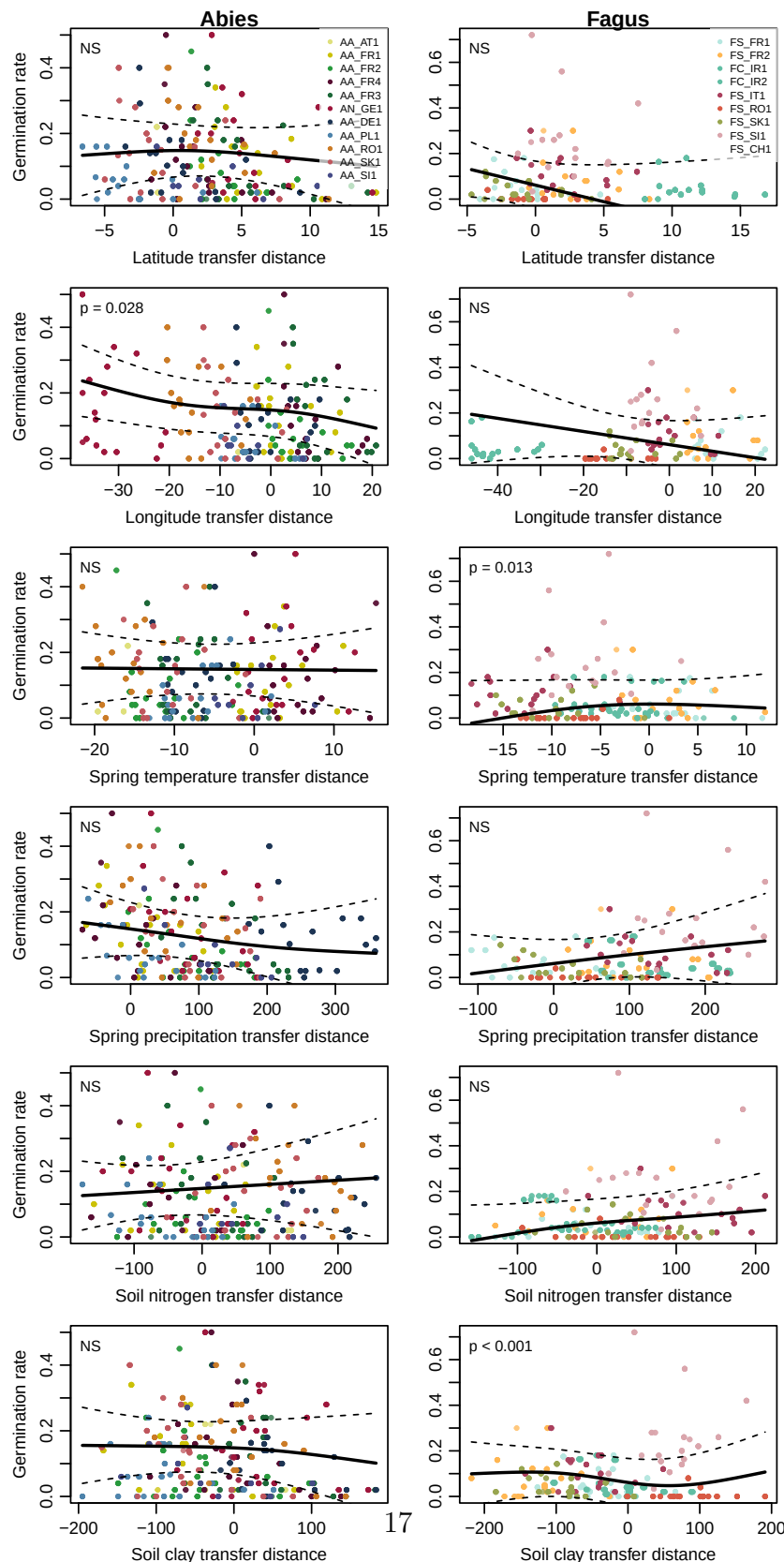


Table S1: Seed-lot information for provenances used in the MyGardenOfTrees pilot experiment. Information was compiled from seed certificates, supplier documentation, catalogs, order correspondence, and personal communications. Abbreviations used: FRM: forest reproductive material, Autoch.: Autochthonous, NS: Not specified, NA: Not available, pers. comm: information was obtained by phone or email, not from the specified Source document, Year: year of seed collection used in this study.

ID	FRM	Origin status	National code	N of seed families	Supplier	Seed quality	Source document	Year
AA_AT1	selected	Autoch.	Ta 15 (3.3/tm: 800–1100 m)	20	Austrian Research Centre for Forests	NA	Austrian OECD certificate	2020
AA_DE1	selected	Autoch.	09182711172	73 (pers. comm.)	Bayerische Staatsforsten AöR – Pflanzgarten Laufen	Germination 49%; thousand-seed weight 45.35 g; moisture content 8.0%	Forest Reproductive Material Master Certificate (Stammzerti- fikat D-09 433 1 4059 20)	2020 (14.09.2020)
AA_FR1	selected	Autoch.	AAL800	20 (pers. comm.)	ONF – Sécherie de la Joux	NA	ONF Forest Reproductive Material Catalogue and order correspondence	2020
AA_FR2	selected	Autoch.	AAL505	20 (pers. comm.)	ONF – Sécherie de la Joux	NA	ONF Forest Reproductive Material Catalogue and order correspondence	2020
AA_FR3	selected	Autoch.	AAL361	20 (pers. comm.)	ONF – Sécherie de la Joux	NA	ONF Forest Reproductive Material Catalogue and order correspondence	2020
AA_FR4	selected	Autoch.	AAL401	20 (pers. comm.)	ONF – Sécherie de la Joux	NA	ONF Forest Reproductive Material Catalogue and order correspondence	2020
AA_HR1	Seed stand	NS	HR-AAL- SS-331/152	NA	Croatian Forest Research Institute	NA	Forematis and order correspon- dence	2020
AA_IT1	NS	NS	LNBS no. 075	Approx. ≥ 10 mother trees, inferred from 630 L cones	Centro Nazionale Biodiversità Forestale di Peri (VR)	Purity 99.6%; viability 24%; thousand-seed weight 40.5 g	Seed availability catalogue (Centro Nazionale Biodiversità Forestale di Peri) and order correspondence	2020

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Table S1: Seed-lot information for provenances used in the MyGardenOfTrees pilot experiment. Continued.

ID	FRM	Origin status	National code	N of seed families	Supplier	Seed quality	Source document	Year
AA_PL1	source- identified	Autoch.	MP/1/41199/19	NA	Leśny Bank Genów Kostrzyca	Viability 42%	Certificate of Origin and Forest Reproductive Material Certificate MR/64203/21/PL	2020
AA_RO1	selected	Autoch.	PS-BR-CV02	At least 20 (Law no. 107/2011)	Ocolul Silvic Brețcu	Purity 94.13%; viability 62.25%; thousand-seed weight 53.36 g; 10,682 viable seeds kg ⁻¹	Forest Reproductive Material Certificate of Origin	2020
AA_SI1	selected	Autoch.	2.0400	At least 25 (pers. comm.)	ZGS / Drevesnica Omorika d.o.o.	Purity 100%; viability 52%; moisture content 10.37%; thousand-seed weight 42.12 g; 12,346 viable seeds kg ⁻¹	Forest Reproductive Material Certificate SI/19-006	2019
AA_SK1	selected	Indigenous	aal216DK-026	At least 20 (pers. comm.)	LESY SR – OZ Semenoles, Liptovský Hrádok	NA	Plant Passport and Master Certificate SK/0085-2021-QVC	2020
AN_GE1	source- identified	NS	HEDEDAN1220 (Ambrolauri Tlugi 12, Georgia)	5	HedeDanmark A/S (HD Seed)	Purity 97.80%; germination 75%; viability 92%; moisture content 7.79%; thousand-seed weight 66.10 g; 13,612 viable seeds kg ⁻¹	Pro forma delivery note and seed analysis certificate	2020
FC_IR1	source- identified (non-EU)	Autoch.	NS	20 (pers. comm.)	Research Institute of Forests and Rangelands (Iran), via Levinsen (Denmark)	NA	Phytosanitary Certificate and collection correspondence	2021
FC_IR2	source- identified (non-EU)	Autoch.	NS	20 (pers. comm.)	Research Institute of Forests and Rangelands (Iran), via Levinsen (Denmark)	NA	Phytosanitary Certificate and collection correspondence	2021
FS_AT1	selected	Autoch.	R.Bu 14 (6.1/hm: 1250–1550 m)	51	Austrian Research Centre for Forests	NA	Austrian OECD certificate	2020
FS_CH1	NS	NS	NS	>100 harvest trees in certified seed stand	Forstbaumschule Josef Kressibucher AG	NA	Nursery invoice and order confirmation	2020

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Table S1: Seed-lot information for provenances used in the MyGardenOfTrees pilot experiment. Continued.

ID	FRM	Origin status	National code	N of seed families	Supplier	Seed quality	Source document	Year
FS.DE1	selected	Autoch.	091810261262	NA	Bayerische Staatsforsten AöR – Pflanzgarten Laufen	Purity 99.84%; germination 81%; thousand-seed weight 226.09 g; moisture content 16.5%	Forest Reproductive Material Master Certificate (Stammzertifikat D-09 433 1 4078 20)	2020
FS.FR1	selected	Autoch.	FSY101	20 (pers. comm.)	ONF – Sécherie de la Joux	NA	ONF Forest Reproductive Material Catalogue and order correspondence	2020
FS.FR2	selected	Autoch.	FSY401	20 (pers. comm.)	ONF – Sécherie de la Joux	NA	ONF Forest Reproductive Material Catalogue and order correspondence	2020
FS.IT1	NS	NS	VEN042	Approx. 3–5 mother trees, inferred from fruit quantity and collection effort	Centro Nazionale Biodiversità Forestale di Peri (VR)	Purity 99.9%; viability 80%; thousand-seed weight 237.4 g	Seed availability catalogue (Centro Nazionale Biodiversità Forestale di Peri) and order correspondence	2020
FS.RO1	selected	Autoch.	NS	At least 20 (Law no. 107/2011)	Ocolul Silvic Teliu	NA	NA	2020
FS.SI1	selected	Autoch.	6.0244	At least 25 (pers. comm.)	ZGS / Drevesnica Omorika d.o.o.	Viability 85.25%; thousand-seed weight 203.64 g; 4,186 viable seedlings kg ⁻¹	Forest Reproductive Material Certificate SI/20-018	2020
FS.SK1	selected	Autoch.	fsy215M-015	At least 20 (pers. comm.)	LESY SR – OZ Semenoles, Liptovský Hrádok	NA	Plant Passport and Master Certificate SK/0057-2021-QVC	2020

Table S2: Environmental covariates used for variable selection in the LASSO regression. Monthly values correspond to 12 variables.

Variable type	Variable full name	Time period	Source database
Micro-garden climate	Chilling days (daily mean temperature < 5 °C) since sowing	Oct 2021-May 2022	CHELSA v2.1 daily
	Growing degree-days (base 5 °C) accumulated until 1 Jul 2022	Jan-Jul 2022	
	Total monthly precipitation	Nov 2021-Jun 2022	
	Mean monthly temperature	Nov 2021-Jun 2022	
Micro-garden geography	Micro-garden elevation	Static	Micro-garden metadata (GPS)
	Micro-garden Latitude		
	Micro-garden Longitude		
Provenance climate	Mean annual number of chilling days (daily mean temperature < 5 °C)	1901-2021	ClimateDT
	Mean annual and monthly temperature		
	Total annual and monthly precipitation		
Provenance geography	Elevation of seed provenance origin	Static	Seed provenance metadata
	Latitude of seed provenance origin		
	Longitude of seed provenance origin		
Provenance and micro-garden soil	Bulk density of the fine earth fraction	Long-term soil property maps at 15-30 cm depth	SoilGrids 2.0
	Cation exchange capacity		
	Coarse fragment volume		
	Clay content		
	Total nitrogen content		
	Organic carbon density		
	Soil pH (H ₂ O)		
	Sand content		
	Silt content		
	Soil organic carbon concentration		

Table S3: Germination metrics

ID	Genus	Treat.	Cycle	G%	t50	GermUncert	CVGTime	GSynchrony
AA_FR1	Abies	dry	long	38	63	3.35	0.26	0.10
AA_FR1	Abies	dry	long	38	63	3.35	0.26	0.10
AA_FR1	Abies	dry	short	26	61	2.78	0.19	0.15
AA_FR1	Abies	joint	long	35	63	3.89	0.28	0.06

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ID	Genus	Treat.	Cycle	G%	t50	GermUncert	CVGTime	GSynchrony
AA_FR1	Abies	joint	short	26	59	3.32	0.21	0.09
AA_FR1	Abies	moist	long	32	61	4.08	0.30	0.04
AA_FR1	Abies	moist	short	25	56	3.54	0.23	0.06
AA_FR2	Abies	dry	long	17	64	2.98	0.31	0.09
AA_FR2	Abies	dry	short	12	58	2.36	0.24	0.15
AA_FR2	Abies	joint	long	26	65	3.81	0.33	0.05
AA_FR2	Abies	joint	short	17	60	3.05	0.26	0.09
AA_FR2	Abies	moist	long	36	65	3.76	0.33	0.06
AA_FR2	Abies	moist	short	22	58	2.98	0.27	0.10
AA_FR3	Abies	dry	short	28	62	3.10	0.15	0.09
AA_FR3	Abies	joint	short	30	65	3.16	0.17	0.10
AA_FR3	Abies	moist	short	31	69	2.75	0.18	0.17
AA_HR1	Abies	dry	short	21	64	3.04	0.24	0.10
AA_HR1	Abies	joint	short	21	63	3.46	0.24	0.06
AA_HR1	Abies	moist	short	21	60	3.04	0.24	0.10
AA_IT1	Abies	dry	long	17	68	3.45	0.30	0.04
AA_IT1	Abies	dry	short	10	59	2.92	0.21	0.04
AA_IT1	Abies	joint	long	17	66	3.62	0.28	0.03
AA_IT1	Abies	joint	short	12	62	2.98	0.21	0.06
AA_IT1	Abies	moist	long	17	66	2.93	0.24	0.10
AA_IT1	Abies	moist	short	14	64	2.55	0.21	0.14
AA_PL1	Abies	dry	short	9	67	2.11	0.19	0.17

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ID	Genus	Treat.	Cycle	G%	t50	GermUncert	CVGTime	GSynchrony
AA_PL1	Abies	joint	short	14	66	2.66	0.13	0.13
AA_PL1	Abies	moist	short	19	65	2.69	0.10	0.13
AA_RO1	Abies	dry	short	35	50	3.57	0.29	0.06
AA_RO1	Abies	joint	short	37	55	3.74	0.28	0.06
AA_RO1	Abies	moist	short	39	57	3.49	0.28	0.08
AA_SK1	Abies	dry	long	32	24	3.40	0.52	0.07
AA_SK1	Abies	dry	short	32	24	3.40	0.52	0.07
AA_SK1	Abies	joint	long	40	28	4.02	0.61	0.05
AA_SK1	Abies	joint	short	37	26	3.77	0.49	0.06
AA_SK1	Abies	moist	long	49	32	4.00	0.60	0.05
AA_SK1	Abies	moist	short	42	28	3.67	0.47	0.07
AN_GE1	Abies	dry	short	64	44	3.97	0.36	0.06
AN_GE1	Abies	joint	short	64	45	4.02	0.34	0.05
AN_GE1	Abies	moist	short	65	45	3.84	0.32	0.07
FC_IR2	Fagus	dry	long	72	44	3.97	0.46	0.06
FC_IR2	Fagus	dry	short	62	39	3.69	0.37	0.07
FC_IR2	Fagus	joint	long	70	47	4.23	0.46	0.05
FC_IR2	Fagus	joint	short	60	44	3.89	0.39	0.06
FC_IR2	Fagus	moist	long	68	48	4.17	0.46	0.05
FC_IR2	Fagus	moist	short	59	45	3.87	0.40	0.06
FS_FR1	Fagus	dry	short	29	60	3.10	0.21	0.10
FS_FR1	Fagus	joint	short	29	60	3.39	0.21	0.08
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ID	Genus	Treat.	Cycle	G%	t50	GermUncert	CVGTime	GSynchrony
FS_FR1	Fagus	moist	short	29	59	3.48	0.22	0.06
FS_SI1	Fagus	dry	short	51	54	3.47	0.26	0.09
FS_SI1	Fagus	joint	short	54	48	3.72	0.28	0.07
FS_SI1	Fagus	moist	short	56	48	3.36	0.30	0.10
FS_SK1	Fagus	dry	short	29	59	2.87	0.29	0.14
FS_SK1	Fagus	joint	short	32	58	3.54	0.31	0.08
FS_SK1	Fagus	moist	short	35	57	3.64	0.33	0.07

Table S4: Wald tests for fixed effects retained in the final mixed-effects germination models for *Abies* and *Fagus*.

Species	Fixed effect	Df	Sum of Sq	Wald statistic	<i>P</i>
<i>Abies</i>	Intercept	1	4.4215	8.0319	0.0046
	Seed weight	1	0.3105	0.5640	0.4527
	Seed moisture content	1	0.1466	0.2664	0.6058
	Garden precipitation spring	1	2.2332	4.0566	0.0440
	Garden temperature spring	1	0.1849	0.3360	0.5622
	Garden soil clay content	1	5.5857	10.1467	0.0014
	Garden soil silt content	1	0.0187	0.0341	0.8536
	Garden soil nitrogen content	1	2.2502	4.0876	0.0432
	Provenance total annual precipitation	1	0.0001	0.0002	0.9877
	Residual (MS)		0.5505		
<i>Fagus</i>	Intercept	1	8.4990	29.2595	< 0.001
	Seed weight	1	5.9773	20.5779	< 0.001
	Seed moisture content	1	0.9239	3.1807	0.0745
	Garden precipitation spring	1	1.8109	6.2345	0.0125
	Garden temperature spring	1	0.3774	1.2994	0.2543
	Garden soil clay content	1	0.7751	2.6685	0.1024
	Garden soil silt content	1	0.5161	1.7768	0.1825
	Garden soil nitrogen content	1	1.6809	5.7868	0.0162
	Provenance total annual precipitation	1	0.0506	0.1743	0.6763
	Residual (MS)		0.2905		

Table S5: Results of generalized additive models (GAMs) relating germination rate estimated from the hidden Markov model (g) to geographic and environmental transfer distances between provenance origin and garden. Temperature and precipitation distances reflect the mean and total for March, April, and May, respectively. Models were fitted separately for *Abies* and *Fagus* and included provenance and garden identity as random effects (random intercepts). The table reports effective degrees of freedom (edf), reference degrees of freedom (Ref. df), F-statistics, and associated P-values for each smooth term. See Figure S11 for a graphical representation.

Genus	Term	edf	Ref. df	F	P
<i>Abies</i>	Longitude distance	2.698	2.935	2.827	0.0278
	Latitude distance	2.314	2.708	2.516	0.1246
	Temperature distance	1.008	1.016	0.006	0.9664
	Precipitation distance	1.970	2.406	0.787	0.3524
	Clay distance	2.164	2.594	1.199	0.2167
	Nitrogen distance	1.000	1.001	0.211	0.6464
	Provenance ID	5.915	10.000	31.999	< 0.001
	Garden ID	13.928	16.000	70.514	< 0.001
<i>Fagus</i>	Longitude distance	1.000	1.000	1.203	0.2732
	Latitude distance	1.000	1.001	2.185	0.1399
	Temperature distance	2.451	2.807	3.763	0.0131
	Precipitation distance	1.473	1.786	0.992	0.4749
	Clay distance	2.868	2.988	7.220	< 0.001
	Nitrogen distance	2.211	2.604	1.074	0.2460
	Provenance ID	5.066	7.000	99.997	< 0.001
	Garden ID	9.795	14.000	19.572	< 0.001
Genus	Adjusted R^2	Deviance explained		REML score	n
<i>Abies</i>	0.703	71.5%		-992.18	775
<i>Fagus</i>	0.683	69.8%		-688.31	560

Methods S2: Climate and soil data and variable selection using LASSO regression

We used climate and soil data to (i) characterize the long-term environments at seed source locations (provenance environments), (ii) describe the environmental conditions experienced by seeds and seedlings at the field sites (micro-garden environments), and (iii) derive thermal indices used to parameterize stage transitions in the Markov model.

Climate variables

First, to characterize the long-term climate at seed source locations, we extracted monthly temperature and precipitation time series from ClimateDT (1 km resolution) [Marchi et al., 2024]. We summarized provenance climate using monthly mean temperature and monthly precipitation totals across the 12 months of the year, and annual temperature and annual precipitation across the period 1901–1980; a baseline period chosen to reduce the influence of recent anthropogenic warming on provenance characterization (Supplementary Data S2). Second, daily temperature and precipitation at field sites were extracted from the CHELSA v2.1 daily product (30 arcsec, ~ 1 km), which provides global daily surface variables including near-surface air temperature and precipitation [Karger et al., 2021, 2017]. We extracted daily data from September 2021 (to cover the pre-sowing period) through October 2022 (covering the first growing season), at the block level within each micro-garden. CHELSA-daily is released as an actively updated dataset and may contain gaps for the most recent period [Karger et al., 2021]. If a single day of daily mean temperature was missing, we gap-filled it using linear interpolation based on the means of the preceding and following days. Blocks of a micro-garden typically fell within the same CHELSA grid cell. However, blocks of micro-gardens 13 and 21 were established in spatially separated locations with contrasting environments. Therefore, climate data were extracted at the block level, and these locations were treated as separate micro-garden units in the mixed-effects analyses. For micro-garden 13, this subdivision is not apparent in the results because the two contrasting environments correspond to the two genus blocks (Table 2).

Third, to model seed-to-seedling and seedling phenological stage transitions, we derived indices from the daily CHELSA data. Growing degree-days (GDD) were calculated as the cumulative sum of daily mean temperature above a base threshold of 5°C , starting on 1 January

of each year. We used this index to parameterize development (“growth”) as a function of accumulated thermal time in the Markov model (see below). For other analysis, we computed a cumulative GDD up to 1 July 2022 (i.e. removing the time aspect) and summarized spring conditions using mean temperature and precipitation sums for selected months during the germination and early establishment period (November 2021–June 2022). Additionally, chilling days were computed as the number of days since sowing with daily mean temperature below 5 °C. For micro-garden 20 (sown on 11 April 2022 due to logistical difficulties), seeds did not experience chilling in the field. To align pre-sowing cold exposure with the other micro-gardens, *Abies* seeds were cold-stratified at 5 °C on moist paper from 19 February 2022 (51 days) and *Fagus* seeds were soaked at 5 °C for 48 hours before sowing.

Soil variables

Finally, soil properties at provenance and micro-garden locations were extracted from SoilGrids (global 250 m soil property maps) [de Sousa et al., 2021]. We used a single depth interval (15–30 cm) to obtain a balanced set of climate-soil predictors while targeting the upper mineral soil layer relevant for early root development and soil water availability relevant for forest regeneration and seedling establishment [Puhlick et al., 2021, Socha et al., 2022, Csilléry et al., 2020b]. The candidate soil predictors were: bulk density, cation exchange capacity, coarse fragments, clay, sand, and silt fractions, soil organic carbon, total nitrogen, organic carbon density, and soil pH in water, extracted for both provenances and micro-gardens.

Variable selection

To reduce the large set of environmental covariates (71 variables, Table S2) in downstream analysis and avoid overfitting, we applied a LASSO regression [Tibshirani, 1996] using the `glmnet` R package [Friedman et al., 2010]. All predictors were standardized prior to the analysis, and the penalization parameter λ was chosen by ten-fold cross-validation (function `cv.glmnet`). We retained 19 (fir) and 13 (beech) variables with non-zero coefficients at $\lambda_{\min}$ as associated with germination (Fig. S10). We then performed clustering of the variables based on their correlations and selected variables relevant to both genera and those that were integrative, such as the mean across several months, rather than monthly variables or those that were easier to interpret. The final set of fixed effects included the seed traits (seed weight and moisture content) and the selected environmental variables.

Methods S3: Description of a hidden Markov model of seed germination, phenological development, and mortality, including the inference and importance sampling scheme

Model description

Let $z_i(t)$ denote the life stage of seed i on day t . We consider three life stages: i) a viable seed that is dormant and has not yet germinated ($z_i(t) = \mathcal{L}_S$), ii) a seed that germinated into a currently growing seedling ($z_i(t) = \mathcal{L}_G$) and iii) non-viable seed or a previously germinated seedling that has died ($z_i(t) = \mathcal{L}_D$). On each day, four events may happen in this order: 1) a seedling will grow, 2) a seedling may die, 3) a seed may germinate and 4) a seed or seedling may be observed. We denote by $t_i^{(g)}$ and $t_i^{(d)}$ the days on which seedling i germinated and died, respectively.

We assume all seedlings can be partitioned into K classes (e.g. combination of provenance and growing location) and let $c_i \in \mathcal{C} = \{\mathcal{C}_1, \dots, \mathcal{C}_K\}$ denote the class of seedling i .

Germination

Let g_c be the probability that a seed of class $c \in \mathcal{C}$ is viable and can germinate. Thus, $z_i(0) = \mathcal{L}_S$ with probability g_{c_i} and $z_i(0) = \mathcal{L}_D$ with probability $1 - g_{c_i}$.

We assume the probability that a seed germinates, i.e. that it transitions from $z_i(t-1) = \mathcal{L}_S$ to $z_i(t) = \mathcal{L}_G$, to depend on the growing degree days $h(t)$ up to day t . We model this dependence using the logistic function

$$\mathbb{P}(z_i(t) = \mathcal{L}_G | z_i(t-1) = \mathcal{L}_S, \mathbf{G}_{c_i}) = \frac{1}{1 + \exp[-\gamma_{c_i}(h(t) - \bar{h}_{c_i})]},$$

where $\mathbf{G}_{c_i} = (g_{c_i}, \gamma_{c_i}, \bar{h}_{c_i})$ denotes the vector of parameters relevant for germination, of which the logistic growth rate γ_{c_i} determines the variation in germination among seeds of a class and $\bar{h}_{c_i}$ the growing degree days necessary to reach a germination probability of $\frac{1}{2}$.

The probability that seed i germinated on day $t_i^{(g)}$ is given by

$$\begin{aligned}
\mathbb{P}(t_i^{(g)} | \mathbf{G}_{c_i}) &= g_{c_i} \mathbb{P}(z_i(t_i^{(g)})) \\
&= \mathcal{L}_{\mathcal{G}} | z_i(t_i^{(g)} - 1) = \mathcal{L}_{\mathcal{S}}, \mathbf{G}_{c_i} \prod_{t=1}^{t_i^{(g)}-1} \mathbb{P}(z_i(t) = \mathcal{L}_{\mathcal{S}} | z_i(t-1) = \mathcal{L}_{\mathcal{S}}, \mathbf{G}_{c_i}),
\end{aligned} \tag{1}$$

where

$$\mathbb{P}(z_i(t) = \mathcal{L}_{\mathcal{S}} | z_i(t-1) = \mathcal{L}_{\mathcal{S}}, \mathbf{G}_{c_i}) = 1 - \mathbb{P}(z_i(t) = \mathcal{L}_{\mathcal{G}} | z_i(t-1) = \mathcal{L}_{\mathcal{S}}, \mathbf{G}_{c_i}).$$

Development

Let $x_i(t)$ quantify the size of a currently growing ($z_i(t) = \mathcal{L}_{\mathcal{G}}$) seedling i on day t . We assume that newly germinated seedlings have a size of zero and grow linearly after germination with class-specific growth rate δ_c such that for a seed i with class c_i

$$x_i(t) = \begin{cases} \delta_{c_i}(t - t_i^{(g)}) & \text{if } z_i(t) = \mathcal{L}_{\mathcal{G}} \text{ and } t_i^{(g)} < t \leq t_i^{(d)} \\ 0 & \text{otherwise} \end{cases}. \tag{2}$$

Note that the variable $x_i(t)$, while referred to as “size”, does not mean an actual measurable size (e.g. tree height), but is used here just as a measure of how a seedling is moving along development stages. Note also that while modeled linearly, we allow for a non-linear translation to life stages (see below).

Death

A currently growing seedling i dies with a probability that is dependent on its current life stage. We assume the highest death rate occurs at germination and then declines exponentially with size. For a seedling i of class c_i we thus have

$$\mathbb{P}(z_i(t) = \mathcal{L}_{\mathcal{D}} | z_i(t-1) = \mathcal{L}_{\mathcal{G}}, \mathbf{D}_{c_i}) = \alpha_{c_i} \exp[-\beta_{c_i} x_i(t)],$$

where $\mathbf{D}_{c_i} = (\alpha_{c_i}, \beta_{c_i})$ denotes the vector of parameters relevant for death and

$$\mathbb{P}(z_i(t) = \mathcal{L}_G | z_i(t-1) = \mathcal{L}_G, \mathbf{D}_{c_i}) = 1 - \mathbb{P}(z_i(t) = \mathcal{L}_D | z_i(t-1) = \mathcal{L}_G, \mathbf{D}_{c_i}).$$

The probability that seed i died on day $t_i^{(d)}$ is given by

$$\mathbb{P}(t_i^{(d)} | t_i^{(g)}, \mathbf{D}_{c_i}) = \prod_{t=t_i^{(g)}+1}^{t_i^{(d)}-1} \left[\mathbb{P}(z_i(t) = \mathcal{L}_G | z_i(t-1) = \mathcal{L}_G, \mathbf{D}_{c_i}) \right] \mathbb{P}(z_i(t_i^{(d)}) = \mathcal{L}_D | z_i(t_i^{(d)}-1) = \mathcal{L}_G, \mathbf{D}_{c_i}). \quad (3)$$

Note that since $x_i(t) = x_i(t-1) + \delta_{c_i}$, these probabilities can be calculated iteratively for efficiency:

$$\begin{aligned} \exp[-\beta_{c_i} x_i(t)] &= \exp[-\beta_{c_i} ((x_i(t-1) + \delta_{c_i}))] \\ &= \exp[-\beta_{c_i} x_i(t-1)] \exp[-\beta_{c_i} \delta_{c_i}]. \end{aligned}$$

This iteration only involves the constant term $\exp[-\beta_{c_i} \delta_{c_i}]$, and no exponentials need to be evaluated.

Emission probabilities: the observed life stages

Let $\tau_1, \dots, \tau_M$ denote M times at which seeds were measured, and let $\tau_0 < \tau_1$ denote the beginning of the experiment. Let $\mathbf{s}_i = (s_i(\tau_1), \dots, s_i(\tau_M))$ denote the life stage identified for seedling i identified at the observation times with $s_i(\tau_m) \in \mathcal{S} = \{\mathcal{S}_0, \mathcal{S}_1, \dots, \mathcal{S}_S\}$. Of these, $\mathcal{S}_0$ denotes the unobserved state, i.e. the seedling has not yet germinated or has already died. For all other stages $\mathcal{S}_s, s > 0$, we model emission probabilities as normalized gamma densities. Let $w_s(x) = \text{Gamma}(x; m_s, \sigma_s^2)$ be the weight of stage s for seedling size x , where $\mathbf{m} = (m_1, \dots, m_S)$ and $\boldsymbol{\sigma}^2 = (\sigma_1^2, \dots, \sigma_S^2)$ are the modes and variances of the gamma distribution relevant for stages $s = 1, \dots, S$. Denoting by $\mathbf{E} = (\mathbf{m}, \boldsymbol{\sigma}^2)$ the vector of emission parameters, the emission probabilities are then given by

$$\mathbb{P}(s_i(\tau_m) = \mathcal{S}_s | z_i(\tau_m), x_i(\tau_m), \mathbf{E}) = \begin{cases} 1 & \text{if } z_i(\tau_m) = \mathcal{L}_{\mathcal{S}} \text{ and } s = 0 \\ 1 & \text{if } z_i(\tau_m) = \mathcal{L}_{\mathcal{D}} \text{ and } s = 0 \\ \frac{w_s(x_i(\tau_m))}{\sum_l w_l(x_i(\tau_m))} & \text{if } z_i(\tau_m) = \mathcal{L}_{\mathcal{G}} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

We assume that the stage $s_{\mathcal{D}} \in \mathcal{S}$ with the highest mortality is known and set $m_{\mathcal{S}_s} = x_{\mathcal{D}}$. Note that the model presented is not identifiable, as multiple combinations of modes m_s and growth rates δ_c can lead to the same emissions. To avoid these non-identifiability issues, we will arbitrarily assume $m_{\mathcal{S}_s} = x_{\mathcal{D}} = 100$.

The emission probability of the full vector of observations is given by

$$\mathbb{P}(\mathbf{s}_i | t_i^{(g)}, \mathbf{z}_i, \delta_{c_i}, \mathbf{E}) = \prod_{m=1}^M \mathbb{P}(s_i(\tau_m) | z_i(\tau_m), \delta_{c_i}(\tau_m - t_i^{(g)}), \mathbf{E})$$

Probability of full realizations

Given the above model, the seeds follow a Markov model with class-specific parameters $\boldsymbol{\theta}_c = (\mathbf{G}_c, \delta_c, \mathbf{D}_c)$ as well as the emission parameters $\mathbf{E}$. For seed i , we distinguish three different types of realizations $\mathbf{z}_i = (z_i(0), \dots, z_i(\tau_M))$ until time τ_M :

1. The seed may be viable, germinate between time points $m - 1$ and m and be observed for the remainder of the experiment at time points $\tau_m, \dots, \tau_M$. We obtain the probability of the observations by integrating out the specific day of germination $\tau_{m-1} \leq t_i^{(g)} < \tau_m$:

$$\mathbb{P}\left(\mathbf{s}_i, z_i(\tau_0, \dots, \tau_{m-1}) = \mathcal{L}_{\mathcal{S}}, z_i(\tau_m, \dots, \tau_M) = \mathcal{L}_{\mathcal{G}} | \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) = \sum_{t^{(g)}=\tau_{m-1}}^{\tau_m} \mathbb{P}(t^{(g)} | \mathbf{G}_{c_i}) \prod_{t=t^{(g)}+1}^{\tau_M} \left[\mathbb{P}(z_i(t) = \mathcal{L}_{\mathcal{G}} | z_i(t-1) = \mathcal{L}_{\mathcal{G}}, \mathbf{D}_{c_i}) \right] \mathbb{P}(\mathbf{s}_i | t^{(g)}, \mathbf{z}_i, \mathbf{E}, \delta_{c_i}),$$

where we used the notation $z_i(t_1, t_2, \dots) = \mathcal{L}$ to indicate that the life stage was $\mathcal{L}$ at all times $t_1, t_2, \dots$ specified.

2. The seed may be viable, germinate between time points $m - 1$ and m , be observed at time points $\tau_m, \dots, \tau_n$ and die prior to time point $\tau_{n+1} \leq \tau_M$. We obtain the probability

of the observations by integrating out the specific days of germination $\tau_{m-1} \leq t_i^{(g)} < \tau_m$ and death $\tau_n < t_i^{(d)} \leq \tau_{n+1}$:

$$\mathbb{P}\left(\mathbf{s}_i, z_i(\tau_0, \dots, \tau_{m-1}) = \mathcal{L}_S, z_i(\tau_m, \dots, \tau_n) = \mathcal{L}_G, z_i(\tau_{n+1}, \dots, \tau_M) = \mathcal{L}_D | \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) = \sum_{t^{(g)}=\tau_{m-1}}^{\tau_m} \mathbb{P}(t^{(g)} | \mathbf{G}_{c_i}) \mathbb{P}(\mathbf{s}_i | t^{(g)}, \mathbf{z}_i, \mathbf{E}, \delta_{c_i}) \sum_{t^{(d)}=\tau_n+1}^{\tau_{n+1}} \mathbb{P}(t^{(d)} | t^{(g)}, \mathbf{D}_{c_i})$$

3. The seed was never observed at any of the time points $\tau_1, \dots, \tau_M$. This may occur due to three distinct reasons:

(a) The seed may be non-viable and stay in state $z_i(t) = \mathcal{L}_D$ for all $t = 0, \dots, \tau_M$. This occurs with probability

$$\mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0, z_i(\tau_1, \dots, \tau_M) = \mathcal{L}_D | \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) = 1 - g_{c_i}$$

(b) The seed may be viable and never germinate. This occurs with probability

$$\mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0, z_i(\tau_1, \dots, \tau_M) = \mathcal{L}_S | \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) = g_c \prod_{t=1}^{\tau_M} \mathbb{P}(z_i(t) = \mathcal{L}_S | z_i(t-1) = \mathcal{L}_S, \mathbf{G}_{c_i})$$

(c) The seed may be viable, germinate between time points $m-1$ and m and die before timepoint m . We obtain the probability of the observations by integrating out the specific days of germination $\tau_{m-1} \leq t_i^{(g)} < \tau_m - 1$ and death $t_i^{(g)} < t_i^{(d)} \leq \tau_m$:

$$\begin{aligned} & \mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0, z_i(\tau_0, \dots, \tau_{m-1}) = \mathcal{L}_S, z_i(\tau_m, \dots, \tau_M) = \mathcal{L}_D | \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) \\ &= \sum_{t^{(g)}=\tau_{m-1}}^{\tau_m-1} \mathbb{P}(t^{(g)} | \mathbf{G}_{c_i}) \sum_{t^{(d)}=t^{(g)}+1}^{\tau_m} \mathbb{P}(t^{(d)} | t^{(g)}, \mathbf{D}_{c_i}). \end{aligned}$$

The probability of $\mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0 | \boldsymbol{\theta}_{c_i}, \mathbf{E}\right)$ is thus given by

$$\begin{aligned}
& \mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0 \mid \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) = \\
& \mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0, z_i(\tau_1, \dots, \tau_M) = \mathcal{L}_{\mathcal{D}} \mid \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) \\
& + \mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0, z_i(\tau_1, \dots, \tau_M) = \mathcal{L}_{\mathcal{S}} \mid \boldsymbol{\theta}_{c_i}, \mathbf{E}\right) \\
& + \sum_{m=1}^{M-1} \mathbb{P}\left(s_i(\tau_1, \dots, \tau_M) = \mathcal{S}_0, z_i(\tau_0, \dots, \tau_{m-1}) = \mathcal{L}_{\mathcal{S}}, z_i(\tau_m, \dots, \tau_M) = \mathcal{L}_{\mathcal{D}} \mid \boldsymbol{\theta}_{c_i}, \mathbf{E}\right)
\end{aligned}$$

Micro-gardens in which seeds were not individually tracked

In our field experiment, seeds were not individually tracked and hence only the counts of life-stages across a set of seedlings of a specific class grown together as one set are available. Let $\mathbf{n}_{j\tau_m} = (n_{j\tau_m 1}, \dots, n_{j\tau_m S})$ denote the vector of stage counts for set j of size $|\mathbf{n}_j|$ at time τ_m and let $\mathbf{n}_j = (\mathbf{n}_{j\tau_1}, \dots, \mathbf{n}_{j\tau_M})$. Let us further denote by $\mathbf{S}$ the matrix of stage assignments to seeds in set j with elements $S_{[im]}$ denotes the stage of seed i at time τ_m . Rows $\mathbf{s}_i$ of $\mathbf{S}$ thus correspond to the observations of seed i . To calculate the probability $\mathbb{P}(\mathbf{n}_j \mid \boldsymbol{\theta}_{c_i}, \mathbf{E})$ of all observations of $\mathbf{n}_j$ of that set across all time points, we need to integrate over all assignments of stages to seeds $\mathbf{S}$:

$$\mathbb{P}(\mathbf{n}_j \mid \boldsymbol{\theta}_{c_i}, \mathbf{E}) = \sum_{\mathbf{S}} \mathbb{P}(\mathbf{n}_j \mid \mathbf{S}) \mathbb{P}(\mathbf{S} \mid \boldsymbol{\theta}_{c_i}, \mathbf{E}),$$

where

$$\mathbb{P}(\mathbf{S} \mid \boldsymbol{\theta}_{c_i}, \mathbf{E}) = \prod_i \mathbb{P}(\mathbf{s}_i \mid \boldsymbol{\theta}_{c_i}, \mathbf{E})$$

and

$$\mathbb{P}(\mathbf{n}_j \mid \mathbf{S}) = \begin{cases} 1 & \text{if } n(\mathbf{S}) = \mathbf{n}_j \\ 0 & \text{otherwise} \end{cases},$$

that is, $\mathbb{P}(\mathbf{n}_j \mid \mathbf{S}) = 1$ if the stage counts per time point $n(\mathbf{S})$ match the observed counts $\mathbf{n}_j$ and zero otherwise.

The space of potential stage assignments $\mathbf{S}$ is very large even for a small number of seeds and cannot be explored in full. We therefore use a Markov approximation to integrate over stage assignments. Since a naive sampling of state assignments $\mathbf{S} \sim \mathbb{P}(\mathbf{S}|\boldsymbol{\theta}_{c_i}, \mathbf{E})$ results in a larger number of samples that are incompatible with the data, we generate samples of $\mathbf{S}$ using importance sampling:

$$\begin{aligned}\mathbb{P}(\mathbf{n}_j|\boldsymbol{\theta}_{c_i}, \mathbf{E}) &\approx \sum_{k=1}^K w_k \mathbb{P}(\mathbf{n}_j|\mathbf{S}_k) \\ \mathbf{S}_k &\sim \mathbb{P}_I(\mathbf{S}|\boldsymbol{\theta}_{c_i}, \mathbf{E}) \\ w_k &= \frac{\mathbb{P}(\mathbf{S}|\boldsymbol{\theta}_{c_i}, \mathbf{E})}{\mathbb{P}_I(\mathbf{S}|\boldsymbol{\theta}_{c_i}, \mathbf{E})}.\end{aligned}$$

We simulate a fraction π of samples under the full model, i.e. by simulating for each seed whether it germinated or not with probability g_c , and if it did germinate, the day of germination according to (1) and the day of death according to (3). These samples have weight $w_k = \frac{1}{\pi}$.

We generate the remaining fraction $1 - \pi$ of samples as follows:

- Let $\tilde{\mathbf{g}}_k$ denote a sorted vector of sampled days at which a seeds germinated. The same day may occur multiple times and if all seeds of a set germinated, its length matches that of the number if seeds of a specific class. Let $\tilde{\mathbf{d}}_k$ analogously be a sorted vector of the intervals (prior to the first, after the last or between consecutive observation times) during which death occurred. We fill these containers as follows: We first identify the minimal number of germination events per interval and pick a random germination day within that interval. If three seeds were observed at time observation time m but four seeds at time $m + 1$, for instance, at least one germination event must have occurred between these observation times and we pick a random day according to (1), truncated to that interval:

$$\mathbb{P}(t_i^{(g)}|\mathbf{G}_{c_i}, \tau_m \leq g < \tau_{m+1}) = \frac{\mathbb{P}(t_i^{(g)}|\mathbf{G}_{c_i})}{\sum_{g'=\tau_m}^{\tau_{m+1}-1} \mathbb{P}(t_i^{(g')}|\mathbf{G}_{c_i})}.$$

We analogously identify the minimal number of death events that occurred in each interval and add those intervals to $\tilde{\mathbf{d}}_k$.

While it is unknown if and when the remaining seeds germinated, the number of additional germination and death events must be equal in each interval. For each of the unaccounted

seeds we therefore i) first simulate if the seed is germinating or not with probability g_c , ii) then, in case the seed germinated, simulate a random germination day (without any interval restriction) according to (1) and add a death event to the interval in which the germination was simulated.

- We next simulate sample pairs of germination dates and death intervals given the constraint that germination must predate death. We start with the earliest death interval $\tilde{d}_{k1}$ of $\tilde{\mathbf{d}}_k$ (the first since the vector is sorted) and randomly pick a germination date from $\tilde{\mathbf{g}}_k$ that is compatible with a death in that interval, i.e. from $\tilde{\mathbf{g}}'_k = (\tilde{g}_{ki} | \tilde{g}_{ki} < \tau_{\tilde{d}_{k1}})$. Once the germination data was simulated, we simulate a valid death date in the chosen death interval that postdates the simulated germination date. We repeat that process for each death interval in sequence.
- We calculate the importance weights w_k as $\frac{1}{1-\pi}$ times the ration between the probability of the simulated $\mathbf{S}_k$ under the full model and its probability under the importance sampling scheme outlined above.

In each iteration, we initially generate 20,000 samples and determine the effective sample size ESS of the samples generated as

$$ESS = \frac{(\sum_k w_k)^2}{\sum_k w_k^2}.$$

We then generate an additional batches of 20,000 samples until the ESS of all samples combined exceeds 2,000.

Implementation

We implemented the above model and inference scheme as a command-line C++ program `tree_growth` using the library `stattools`. The implementation, along with a brief user manual, is available at https://bitbucket.org/wegmannlab/tree_growth/. All estimations in this paper were done with commit `a9ae485`.